When Not to Think

Mitigating the Overthinking Effect in Content Moderation

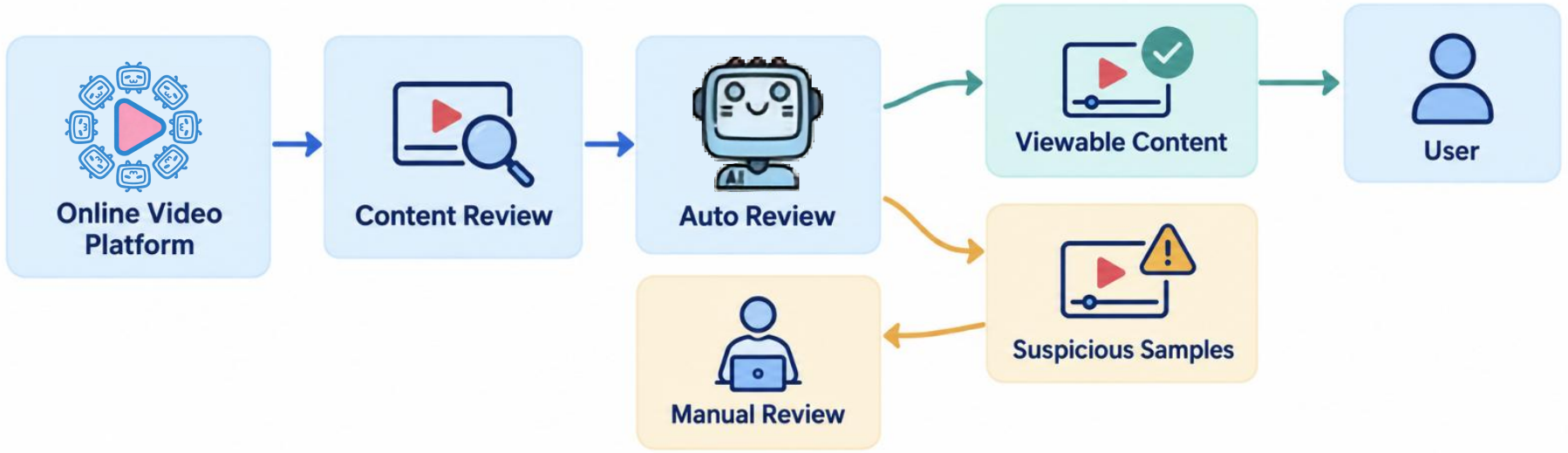
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Content Moderation



SCALE

Screen millions of submissions and focus human review on cases that need attention.

CONTEXTUAL AMBIGUITY

Similar words or images can be benign or violating depending on context.
Over-interpretation can trigger false alarms.

When to Think

- Context-aware analysis, without over-interpretation.



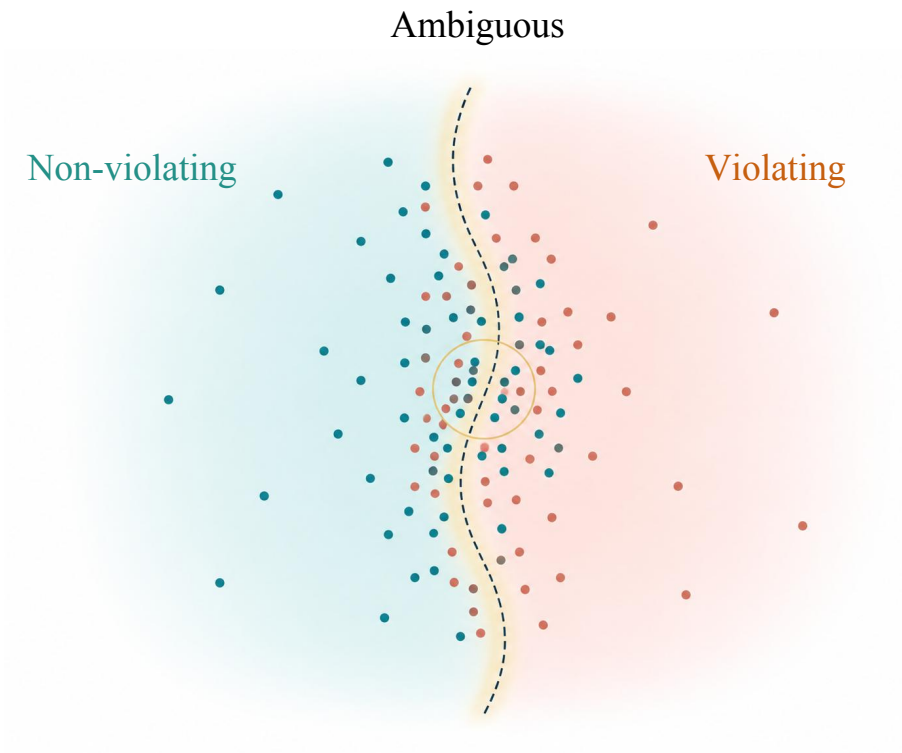
Input. A clip from a video-game livestream of the Soulslike game *Where Winds Meet*.

CoT-off: Non-Violating **CoT-on:** Gore Game

Reasoning excerpt.
"*The video contains an obvious game UI . . . identified as game footage . . . The cover directly shows the boss 'executing' the player character, accompanied by a blood-splatter effect . . . Conclusion: violated.*"

Research Motivation

Boundary-dense classification: *Ambiguous, borderline samples with annotator disagreement.*



Mathematics or Symbolic Reasoning

A well-defined answer and a relatively clear solution path.
Reasoning can help the model find that answer,
or it may simply add computational cost.

Content moderation

Ambiguous comments may contain quotations,
sarcasm, lyrics, sensitive terms, and multiple
meanings, leaving no single correct reasoning path.

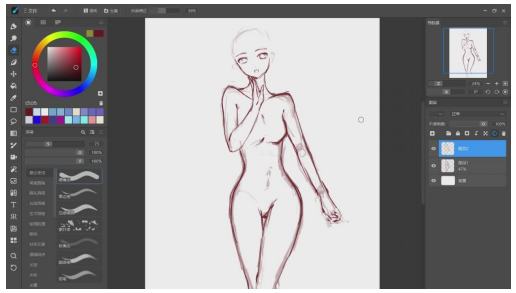
Potential Risk

Extended reasoning can amplify weak cues
into unsupported violation judgments.

Unnecessary CoT harm prediction quality, rather than merely waste computation !!

The Overthinking Effect

CASE 1: Image (Digital Drawing Screenshot)



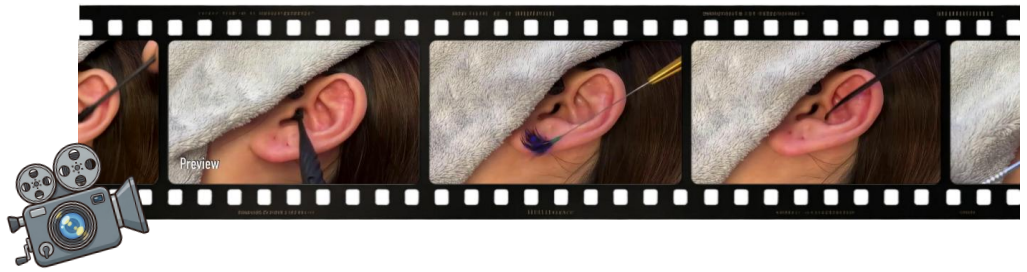
Screenshot (A drawing app with an anime line-art sketch)

Reasoning-off	Reasoning-on
Direct Verdict: No violation	Exposed genitalia; Female nudity



Scene Type: Digital-drawing software interface
Content Identification: At the center of the canvas is a hand-drawn anime female character line sketch. The character is fully nude; At the lower body the lines clearly outline the structure of the female genitals. The chest area has outline lines only, with no nipples drawn.
Rule Match: Exposed genitalia; Female nudity;

CASE 2: Video (Ear-cleaning ASMR Clip)



Video Clip (various ear-pick tools shown, but no ear wax actually removed)

Reasoning-off	Reasoning-on
Direct Verdict: No violation	Disgust-relief

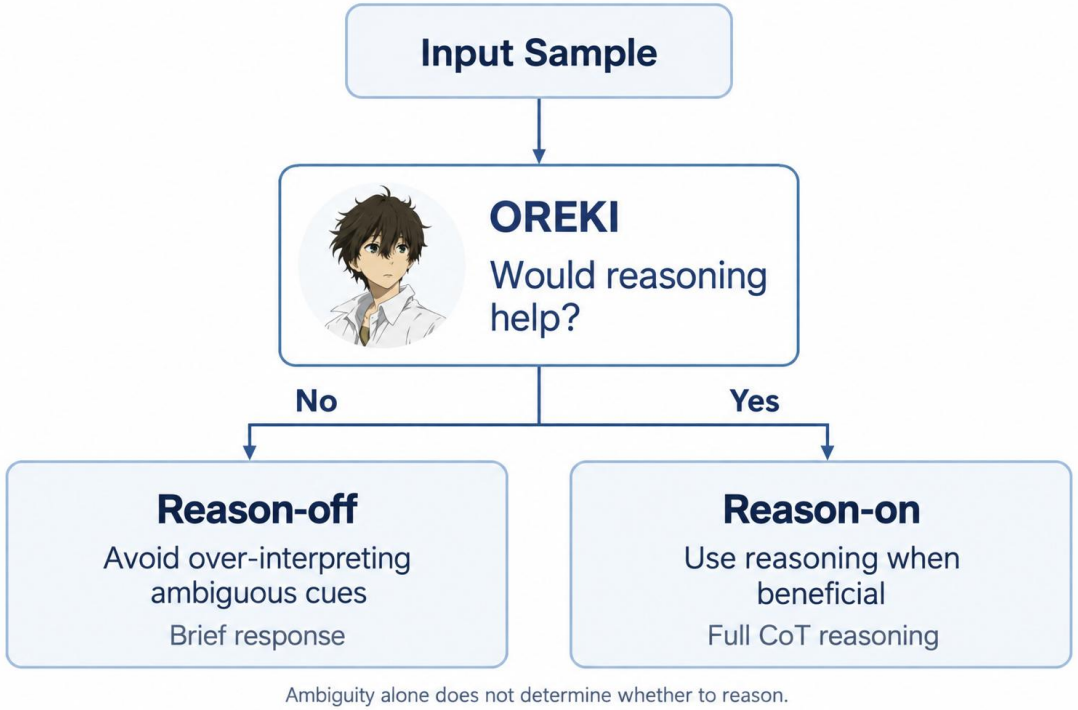


Scene Type: Real footage
Content Identification: Cover shows a close-up of an ear-pick tool cleaning inside the ear canal. Following frames show the ear-cleaning process throughout, using various tools (feather stick, ear pick, etc.). Clearly a wax-digging scene.
Rule Match: Disgust-relief. The frames clearly and directly show the wax-digging process.

OREKI: When to Reason

「 *OREKI： Named after Houtarou Oreki "I don't do anything unnecessary."* 」





MODERATION ANALOGY

Clearly safe

Passes instantly

High-risk

Handled promptly

Ambiguous

Skip CoT to avoid false alarms

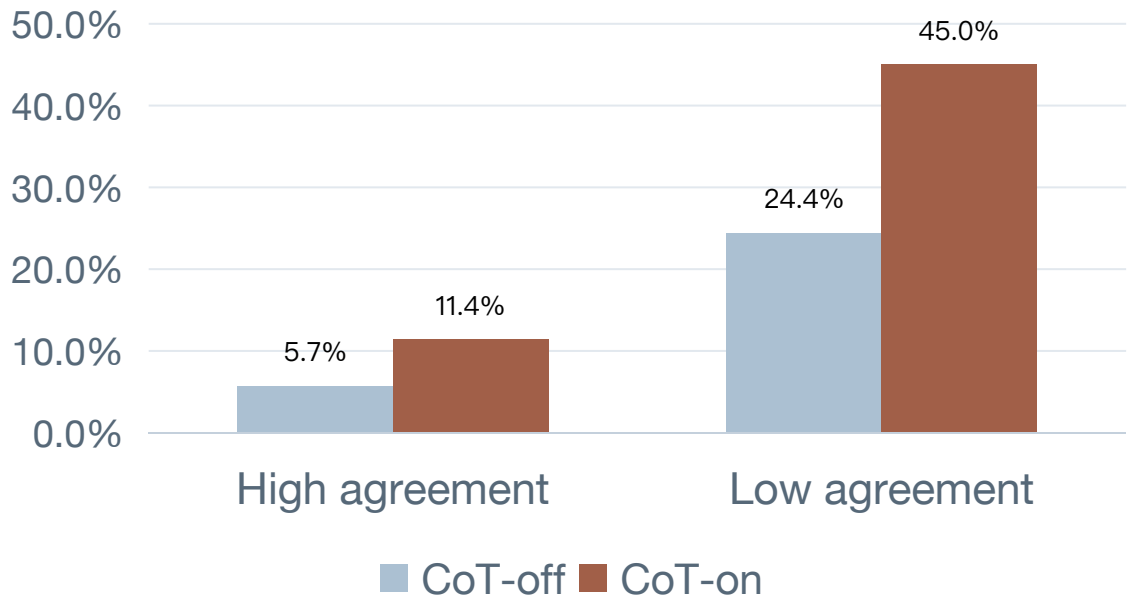
Complex

Use CoT for better discrimination

CoT raises false positives before fine-tuning

Qwen3.5-4B on CivilComments · Binary toxicity detection · No fine-tuning

False-positive rate on non-toxic samples



Larger absolute FPR increase on ambiguous inputs: +20.6 pp

LOW-AGREEMENT TOXIC RECALL

38.7% to 65.1%

+26.4 percentage points

OVERALL F1

0.494 to 0.607

Overall improvement hides a cost

Higher overall F1 can mask more false alarms on benign content

OREKI: A Two-Stage Pipeline

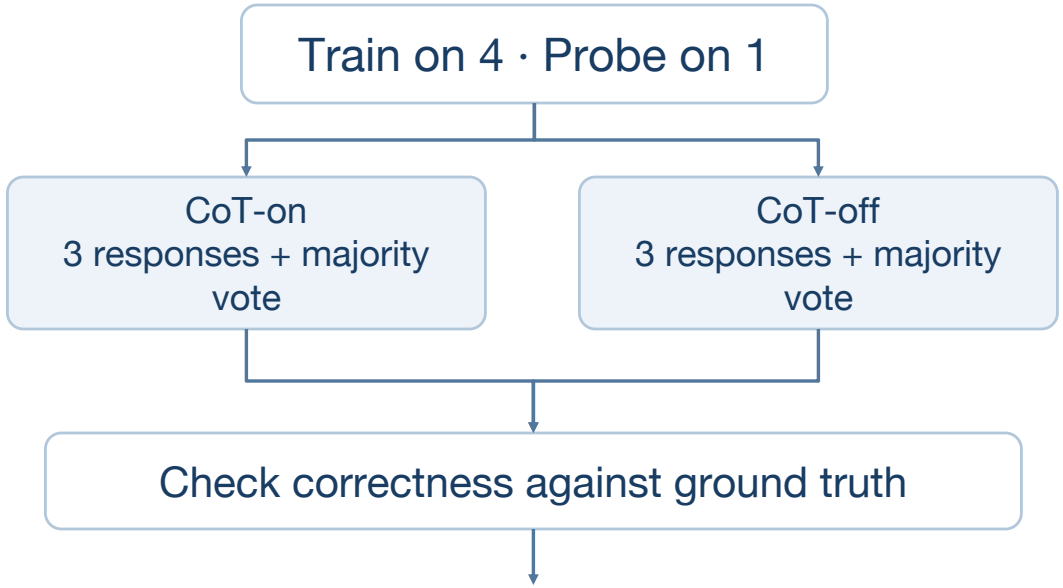
01 CROSS-PROBING

Five-fold necessity labeling

TRAIN ON FOUR FOLDS | HOLD OUT

Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5

Rotate the held-out fold across five rounds



Output: cot / no-cot labels

02 FINE-TUNING

Mixed-data training

c_i^{cot}	$c_i^{\text{no-cot}}$	Label
✓	✓	`no-cot` (unnecessary)
✓	×	`cot` (beneficial)
×	✓	`no-cot` (harmful)
×	×	`cot` (give CoT a chance)

Cot Targets

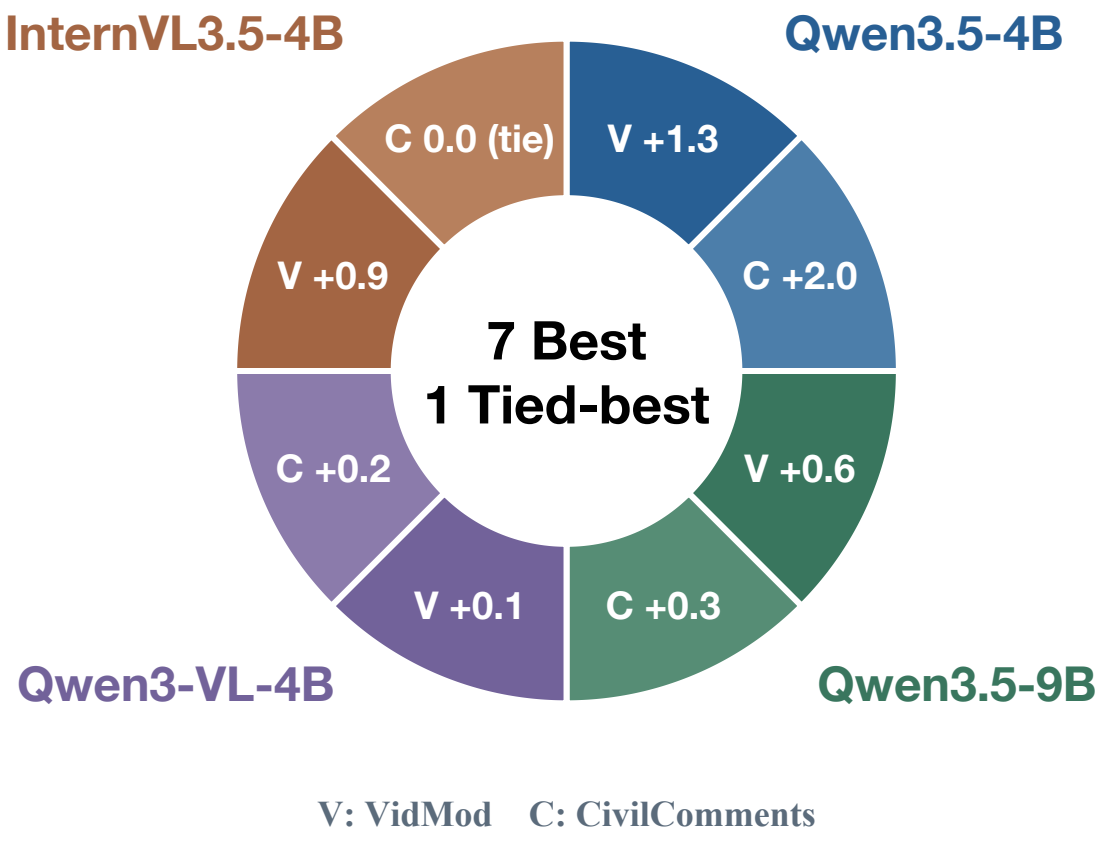
Retain the full reasoning chain.

No-cot Targets

Use a brief meta-reasoning template followed by the answer.

OREKI improves F1 on both tasks

Overall F1 after supervised fine-tuning



V: VidMod C: CivilComments

Consistent across all 8 settings

OREKI matches or exceeds both fixed strategies across four backbones and two tasks.

Why selective reasoning helps

Always-CoT can over-interpret benign cues.
Never-CoT can forgo useful reasoning.
OREKI learns which samples need CoT.

The benefit varies by model

Overall F1 is consistently competitive, while subgroup tradeoffs remain.

Gain vs. the stronger fixed strategy, in F1 points (0–100). Equal sectors represent settings, not gain size. Source: Table 3.

Adaptive reasoning combines both strengths

CivilComments · Qwen3.5-4B · Low-agreement samples



Compared with Always-CoT

–21.0 pp FPR

with only 0.8 pp lower recall

Compared with Never-CoT

–3.0 pp FPR

with 3.3 pp higher recall

Learn when reasoning helps

OREKI selects CoT per sample, retaining useful reasoning while limiting over-interpretation of benign cues.

Near Always-CoT recall, with FPR below Never-CoT

Why Selective Routing Matters

Method	CoT rate	F1	LA-V R↑	LA-NV FPR↓
Always-CoT	100%	.792	.820	.467
Never-CoT	0%	.831	.779	.287
Rate-matched random	12.88%	.825	.784	.310
Lightweight router (F1)	50.95%	.810	.812	.398
Lightweight router (rate)	13.77%	.827	.789	.314
OREKI	12.88%	**.851**	.812	**.257**

LA-V: low-agreement violating samples
LA-NV: low-agreement non-violating samples

Selection matters beyond frequency

Matching the CoT rate controls for how often reasoning is used.
The random baseline isolates the value of choosing which inputs receive it.

The decision belongs inside the model

The router uses the same probing labels.
Its gap to OREKI supports learning the routing decision jointly with the response.

Reasoning benefit is sample-dependent

CoT can recover ambiguous violations, yet over-interpret benign inputs.
Selective use addresses this asymmetry.

At the same 12.88% CoT rate, OREKI improves F1 from 0.825 to 0.851.

Production impact at platform scale

Full moderation stream · Millions of submissions per day

HUMAN REVIEW

Higher confirmed-violation share

+22.7 pp

in the human-review queue

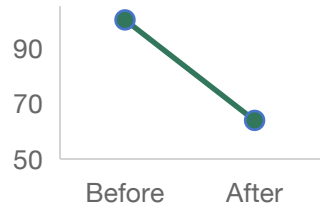
A larger share of auto-flagged items
is confirmed as genuinely violating.
Review effort focuses on real problems.

USER FEEDBACK

Lower negative-feedback rates

Category 1

–36.2%



0.047% → 0.030%

Category 2

–39.3%



0.028% → 0.017%

Category 3

–24.0%



0.005% → 0.0038%

Relative reductions; slopes use a shared index (before = 100).

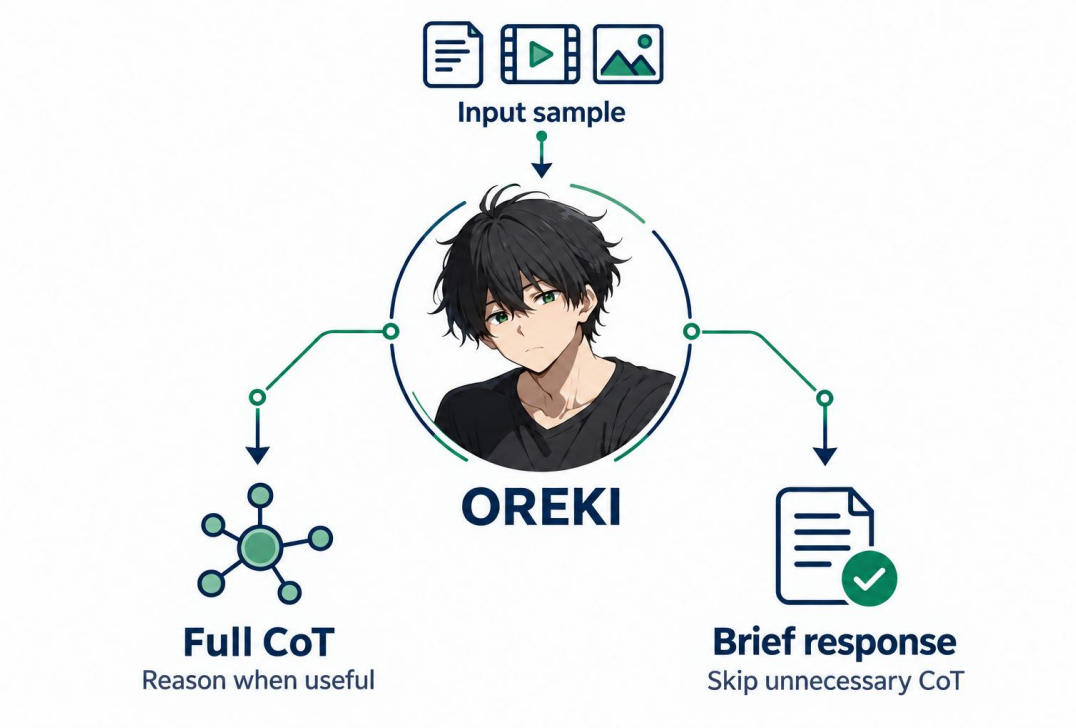
Conclusion

Reliable moderation requires learning when not to think.

THE OVERTHINKING EFFECT

Unnecessary CoT can inflate false alarms on ambiguous non-violating content.

OREKI learns when reasoning helps each model.



Best or tied-best F1 across four backbones and both tasks

Thanks for your attention.

Bilibili Inc., Shanghai, China



